CoRS: Consistency-of-Reasoning Selection Literature Review

VLA Project, Week 2

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1 Adeniyi Onaolapo: Detailed literature review

1.1 Alpamayo-R1: Bridging Reasoning and Action Prediction (arXiv 2511.00088), plus the Hugging Face model card.

- **What they did:** Introduced a vision-language-action model for autonomous driving that combines chain-of-causation reasoning with trajectory prediction, targeting long-tail driving scenarios.
- **What they did not do:** The paper is not a training-free, test-time method for selecting among multiple sampled reasoning–trajectory pairs using an agreement score.
- **How CoRS differs:** CoRS is a model-agnostic inference-time selection and confidence approach; it evaluates whether each candidate's stated intent matches its predicted motion, then selects a consensus trajectory.

Paper: `https://arxiv.org/abs/2511.00088` *Model card:* `https://huggingface.co/nvidia/Alpamayo-R1-10B`

1.2 Is VLA Reasoning Faithful? Probing Chain-of-Causation in Driving Models

- **What they did:** Probed Alpamayo-R1's Chain-of-Causation faithfulness over 300 inferences using entity/action fidelity, consistency checks, and visual perturbations; proposed SafeDriveX.
- **What they did not do:** Test other models, validate with a VLM-based extractor, or build or evaluate SafeDriveX; entity matching relied on keyword rules.
- **How CoRS differs:** The paper doesn't mention CoRS. If it selects among sampled outputs by consistency, that acts at inference, whereas this paper only diagnoses and proposes RL fine-tuning.

Paper: `https://arxiv.org/abs/2605.17268`

1.3 Lost in Fog: Sensor Perturbations Expose Reasoning Fragility in Driving VLAs

- **What they did:** Perturbed Alpamayo R1's camera inputs (noise, lighting, fog) across 1,996 clips, linking changed explanations to large trajectory errors.
- **What they did not do:** Closed-loop testing, gradient-based attacks, other architectures, a deployable monitor (no clean reference), or proof of causation.
- **How CoRS differs:** The paper never mentions CoRS. CoRS is a runtime selection method—a perturbation study. Its optional augmentation stability score could be compared with this work.

Paper: `https://arxiv.org/abs/2605.21446`

1.4 VLADriveBench: Evaluating CoT-Action Relationship in VLA

- **What they did:** Built VLADriveBench, combining CoT quality metrics, observational alignment, and CoT-injection interventions (validated by selfsplice) on ORION and Alpamayo in CARLA.
- **What they did not do:** Evaluate in-distribution for Alpamayo, test other CoT designs (visual/counterfactual), or characterize salience thresholds, or propose any method to improve CoT-action consistency.

- **How CoRS differs:** VLADriveBench manipulates the reasoning trace to test whether it matters. A benchmark evaluates systems; CoRS is a test-time method. CoRS's reasoning–action agreement rate and "says-stop-but-moves" metric may be relevant evaluation measures.

Paper: `https://arxiv.org/abs/2606.12706`

1.5 Do What You Say: Runtime Reasoning-Action Alignment Verification

- **What they did:** Their method, SEAL, samples 10 action sequences per text plan, simulates the outcomes, and has GPT-4o pick the one matching the plan. It needs no retraining.
- **What they did not do:** They ran no real-robot tests, used no learned world model, and did not fix weak base-policy proposals or the verifier's wrist-view and occlusion errors.
- **How CoRS differs:** SEAL verifies visually, by checking whether the simulated outcome looks like what the plan says. CoRS, verifies symbolically, by converting the text to discrete intent labels, which would then be matched against the trajectory.

Paper: `https://arxiv.org/abs/2510.16281`

1.6 DriveVer: Test-Time Verifier for Autonomous Driving

- **What they did:** DriveVer is a 34M-parameter plug-in that scores one planned trajectory from images and, when flagged, geometrically refines it, lifting NAVSIM PDMS.
- **What they did not do:** They didn't sample multiple candidates at test time, check reasoning-trajectory consistency, use text reasoning, or evaluate closed-loop or on real vehicles.
- **How CoRS differs:** DriveVer is a trained corrector of a single output. CoRS is a training-free selector among many, using internal consistency instead of an external learned judge.

Paper: `https://arxiv.org/abs/2607.00399`

1.7 Self-Consistency Improves Chain-of-Thought Reasoning

- **What they did:** Proposed self-consistency: sample multiple reasoning paths and use agreement among their answers to improve reasoning performance.
- **What they did not do:** It does not, in its general form, verify that generated reasoning is faithful to a continuous physical trajectory or select a driving trajectory medoid.
- **How CoRS differs:** CoRS applies a related sampling-and-consensus idea to driving, adds an explicit text–motion agreement score, and selects among trajectories rather than only aggregating discrete answers.

Paper: `https://arxiv.org/abs/2203.11171`

1.8 On the Role of Faithfulness in Embodied Reasoning

- **What they did:** Built Pinocchio, a critic scoring consistency across the reasoning chain, and used it as a dense GRPO reward for a driving VLA.
- **What they didn't do:** Prove mechanistic faithfulness, validate in closed-loop, or avoid Gemini label noise. Consistency is necessary, not sufficient.

- **How CoRS differs:** Pinocchio changes the policy through training; CoRS leaves the policy untouched and filters its outputs at inference. CoRS operationalizes one form of faithfulness at inference time: whether a candidate's stated driving intent agrees with the motion in its trajectory.

Paper: `https://arxiv.org/abs/2607.04681`

A Owen R. notes

A.1 Alpamayo-R1: Bridging Reasoning and Action Prediction

- **What they did:** Built a driving VLA that combines reasoning and trajectory prediction, using training to improve their consistency.
- **What they did NOT do:** Select among multiple outputs using training free, agreement weighted consensus.
- **How CoRS differs:** CoRS checks text–motion agreement and selects a consensus trajectory without retraining.

A.2 Is VLA Reasoning Faithful? Probing Chain-of-Causation in Driving Models

- **What they did:** Analyzed 300 outputs and found approximately 48% reasoning–action consistency.
- **What they did NOT do:** Implement and evaluate a method for selecting better trajectories.
- **How CoRS differs:** CoRS uses agreement to select candidates and tests whether this improves accuracy.

A.3 Lost in Fog: Sensor Perturbations Expose Reasoning Fragility in Driving VLAs

- **What they did:** Tested image perturbations and found explanation changes associated with much larger trajectory deviations.
- **What they did NOT do:** Use reasoning stability to select better trajectories or establish causation.
- **How CoRS differs:** CoRS selects candidates and can include reasoning stability in its confidence score.

A.4 VLADriveBench: Evaluating CoT-Action Relationship in VLA

- **What they did:** Created a benchmark testing reasoning quality, action alignment, and reasoning's influence on actions.
- **What they did NOT do:** Introduce an agreement weighted trajectory selection method.
- **How CoRS differs:** CoRS uses agreement to choose trajectories, without claiming to prove causal faithfulness.

A.5 Do What You Say: Runtime Reasoning-Action Alignment Verification

- **What they did:** Sampled robotic actions, simulated outcomes, and used a vision-language model to select the best plan match.
- **What they did NOT do:** Provide a simulator free selection method for driving.
- **How CoRS differs:** CoRS directly compares text and trajectory intent, then applies consensus and medoid selection.

A.6 DriveVer: Test-Time Verifier for Autonomous Driving

- **What they did:** Built a learned verifier that scores driving trajectories and predicts geometric corrections.
- **What they did NOT do:** Use written reasoning to check intent–motion agreement.
- **How CoRS differs:** CoRS selects existing trajectories using agreement and consensus without training another verifier.

A.7 Self-Consistency Improves Chain-of-Thought Reasoning

- **What they did:** Improved reasoning performance by sampling multiple reasoning paths and aggregating their answers.
- **What they did NOT do:** Check reasoning against physical motion or select driving trajectories.
- **How CoRS differs:** CoRS adds text–motion agreement weighting and selects a representative trajectory.

A.8 On the Role of Faithfulness in Embodied Reasoning

- **What they did:** Developed Pinocchio, a learned reasoning critic, and used it for reinforcement learning post training.
- **What they did NOT do:** Provide a training free consensus selector or prove mechanistic faithfulness.
- **How CoRS differs:** CoRS keeps the policy unchanged and selects outputs using stated intent and motion agreement.

B Zach Fogle notes

B.1 Alpamayo-R1

- **What they did:** Built a driving VLA that generates reasoning text and a future trajectory.
- **What they did NOT do:** They did not sample many outputs and use reasoning-action agreement to select one.
- **How we differ:** CoRS uses Alpamayo outputs and picks the most consistent reasoning-trajectory pair at test time.

B.2 Is VLA Reasoning Faithful?

- **What they did:** Showed that Alpamayo's reasoning often does not match its actual trajectory.
- **What they did NOT do:** They only measured the problem and did not try to correct it.
- **How we differ:** CoRS uses the disagreement itself to help select a better trajectory.

B.3 Lost in Fog

- **What they did:** Showed that small image changes can change the reasoning and cause much larger trajectory errors.
- **What they did NOT do:** They did not use reasoning stability to improve trajectory selection.
- **How we differ:** CoRS can use reasoning stability as part of its confidence score.

B.4 VLADriveBench

- **What they did:** Created a benchmark for measuring how well VLA reasoning matches driving actions.
- **What they did NOT do:** It is only an evaluation method and does not select better trajectories.
- **How we differ:** CoRS uses reasoning-action agreement during inference to choose an output.

B.5 Do What You Say

- **What they did:** Used reasoning-action verification at runtime to choose better actions for robotic manipulation.
- **What they did NOT do:** It was not designed for autonomous driving and uses an extra verifier.
- **How we differ:** CoRS works directly with driving reasoning and trajectories without requiring another trained verifier.

B.6 DriveVer

- **What they did:** Built a test-time verifier that scores and improves driving trajectories.
- **What they did NOT do:** It does not use the model's reasoning text.
- **How we differ:** CoRS uses reasoning-action consistency instead of a separately trained trajectory verifier.

B.7 Self-Consistency Improves Chain-of-Thought Reasoning

- **What they did:** Sampled multiple reasoning paths and selected the answer supported by the strongest consensus.
- **What they did NOT do:** It was made for language reasoning, not physical actions or driving.
- **How we differ:** CoRS applies the same idea to multiple reasoning-trajectory pairs.

B.8 On the Role of Faithfulness in Embodied Reasoning

- **What they did:** Studied whether a model's reasoning actually reflects the actions it takes.
- **What they did NOT do:** Their method requires additional training and a learned critic.
- **How we differ:** CoRS is training-free and checks reasoning-action agreement directly at runtime.

C Rohit Vummadi: Expanded Week 2 summaries

These summaries compare eight papers with our proposed Consistency-of-Reasoning Selection (CoRS) method. CoRS aims to select among a frozen model's reasoning–trajectory pairs using agreement-weighted intent voting and a medoid, or the most central existing trajectory. Its benefits have not yet been established.

C.1 Alpamayo-R1: Bridging Reasoning and Action Prediction

arXiv:2511.00088

- **What they did:** Developed a driving VLA that generates Chain-of-Causation explanations and future trajectories. They combined a vision-language backbone, an action expert, and supervised and reinforcement learning. Their experiments showed improvements in trajectory prediction and reasoning–action consistency.
- **What they did not do:** They did not evaluate a training-free selector that combines agreement-weighted intent voting with medoid selection. The paper's reinforcement-learning results also should not be assumed to apply to the original public supervised-learning checkpoint.
- **How we differ:** We plan to keep the model's weights fixed and select among its sampled outputs. Our experiment tests whether reasoning agreement adds value beyond selecting trajectories using motion alone.

C.2 Is VLA Reasoning Faithful?

arXiv:2605.17268

- **What they did:** Evaluated 300 Alpamayo-R1 inferences across 100 driving clips, checking scene descriptions, stated actions, and predicted motion. They reported a mean reasoning–action consistency score of 48.3%, with 37.9% of stop-claimed cases failing their stopping check.
- **What they did not do:** They did not demonstrate a working candidate-selection method that corrects these mismatches. The findings come from one model and depend on keyword extraction, annotations, and motion thresholds.
- **How we differ:** We propose using text–trajectory agreement to choose a candidate, rather than only measuring disagreement. We will evaluate trajectory error and intent accuracy alongside agreement to check whether selection actually improves the output.

C.3 Lost in Fog: Sensor Perturbations Expose Reasoning Fragility

arXiv:2605.21446

- **What they did:** Tested Alpamayo on 1,996 driving scenarios under eight image corruptions. When explanations changed, deviation between clean and perturbed predicted trajectories was about 5.3 times greater. This measures prediction instability, not a fivefold increase in error against ground truth.
- **What they did not do:** They did not use reasoning agreement to select a better trajectory. Their monitoring experiment also required paired clean and corrupted inputs, so it did not establish a deployable monitor without a clean reference.
- **How we differ:** We plan to select among sampled reasoning–trajectory pairs. We may additionally use intent stability under mild image changes as a confidence signal, then evaluate whether that confidence identifies failures.

C.4 VLADriveBench: Evaluating the CoT–Action Relationship

arXiv:2606.12706

- **What they did:** Evaluated reasoning quality, action alignment, and the effects of replacing reasoning text in three driving models. They showed that high observed agreement does not necessarily mean reasoning semantics cause the action, while lower agreement can coexist with strong causal influence.
- **What they did not do:** They did not introduce a CoRS-style selector. Their controlled CARLA experiments diagnose reasoning–action relationships, and CARLA is outside the Alpamayo models' real-world training distribution.
- **How we differ:** We propose selecting outputs using agreement without intervening on their reasoning. We will treat agreement as a selection signal, not proof that the reasoning is causally faithful or that the action is correct.

C.5 Do What You Say: Runtime Reasoning–Action Alignment Verification

arXiv:2510.16281

- **What they did:** Introduced SEAL, which samples robot action sequences, simulates their outcomes, and uses a pretrained vision-language verifier to check whether they fulfill a textual plan. This runtime steering improved manipulation performance, including under unfamiliar conditions.
- **What they did not do:** They did not demonstrate simulator-free driving selection based on trajectory-derived intent. Their runtime selection is training-free, but the experimental system includes a trained reasoning policy and depends on simulated outcomes and a verifier.
- **How we differ:** We plan to compare driving intent extracted directly from text and motion, then apply weighted voting and medoid selection. Our proposed selector does not require simulator rollouts or an additional learned verifier.

C.6 DriveVer: Test-Time Verifier for Autonomous Driving

arXiv:2607.00399

- **What they did:** Trained a compact verifier that uses images, vehicle state, navigation commands, and an initial trajectory to predict whether correction is needed and how to adjust it. It improved several planners on NAVSIM, including DiffusionDrive from 88.1 to 89.0 PDMS.

- **What they did not do:** They did not use reasoning text to guide verification. Although applied at test time without retraining the base planner, DriveVer requires training its own verifier and changes the trajectory rather than only selecting one.
- **How we differ:** We propose choosing an existing candidate using reasoning–motion agreement. We will not train a correction model or modify the selected trajectory's coordinates.

C.7 Self-Consistency Improves Chain of Thought Reasoning

arXiv:2203.11171

- **What they did:** Sampled multiple reasoning paths from one language model and selected the most frequent final answer. This training-free voting approach improved reasoning benchmarks; for example, PaLM-540B's GSM8K accuracy increased from 56.5% to 74.4%.
- **What they did not do:** They did not evaluate driving trajectories or agreement between language and physical motion. Their voting operates on identifiable answers, while continuous trajectories require a definition of which outputs count as equivalent.
- **How we differ:** We propose grouping candidates by driving intent, weighting votes by text–motion agreement, and selecting a medoid within the winning group. Consensus alone will not be treated as evidence of correctness.

C.8 On the Role of Faithfulness in Embodied Reasoning

arXiv:2607.04681

- **What they did:** Developed Pinocchio, a learned critic that checks consistency among observations, scene descriptions, justifications, meta-actions, and trajectories. They used its scores during reinforcement learning to improve overall reasoning consistency, with a modest tradeoff in trajectory accuracy.
- **What they did not do:** They did not provide a training-free selector over frozen-model candidates. Their consistency checks also serve as a behavioral approximation, not proof that the reasoning causally determines the action.
- **How we differ:** We propose selection without critic or policy training. Our simpler intent-agreement check does not assess every intermediate reasoning step, so we must test its limits and whether it improves held-out trajectory outcomes.

D Isaac R.: Week 2 literature review notes

D.1 Alpamayo

- **What they did:** Train a general CLM into a policy for driving by adding an action modality, With teaching to explain via stage 2.
- **What they did not do:** test if the reasoning texts have a selection indicator, not verified.
- **How CoRS differs:** We put a layer on top of Alpamayo to not induce the issue of style relabeling, gradients.

D.2 Faithful VLA

- **What they did:** Display low fidelity correlates with higher ADE and most trajectories are fragile under perturbation.
- **What they did not do:** have an implementation, just a diagnosis of a proposed safety layer.
- **How CoRS differs:** CoRS contains an actionable selector with scalar agreement scores.

D.3 Lost in Fog

- **What they did:** Show CoC generation is associated with lower ADE, with the input defenses not giving significant gain.
- **What they did not do:** Resample or select candidates, with just a proposition to fix the issue.
- **How CoRS differs:** We use reasoning stability as input to selection, not just a warm warning.

D.4 VLADriveBench

- **What they did:** Show alignment and causal influence can diverge, Alpamayo looks less aligned but semantically causal.
- **What they did not do:** Test whether alignment weighted acquisition is better than trajectory only aggregation.
- **How CoRS differs:** CoRS has the trajectory only baseline comparison that VLADriveBench did not run.

D.5 SEAL

- **What they did:** Show plan to action verification provides identification gains, OOD robustness with VLM verification being the bottleneck.
- **What they did not do:** Have a structured intent space, which means there is no group consensus.
- **How CoRS differs:** Replaces the VLM verifier with a geometric intent extractor and parser, providing a valuable and auditable method.

D.6 DriveVer

- **What they did:** Showed with a cosine loss beating predicting absolute residuals, directional refinement.
- **What they did not do:** Provide a language channel for the verifier, so it cannot check the validity of the model's actions.
- **How CoRS differs:** You can ask if reasoning adds value over a geometric selector, which a verifier only design can't do.

D.7 Self-consistency

- **What they did:** Display as a mediator between open ended generation and fixed tasks labelled as a decoding strategy.
- **What they did not do:** Contain continuous or geometric output answer sets.
- **How CoRS differs:** CoRS weighs pairs by agreement and trajectory space.

D.8 Pinocchio

- **What they did:** Have distinct functional and faithful reasoning split, with faithfulness having five edges over its reasoning graph.
- **What they did not do:** The model contains a change in weights, with consistency being a reward.
- **How CoRS differs:** CoRS keeps the same policy throughout.